

ICMP firewall

Write a PyDiver-based firewall that can block either incoming or outgoing ICMP ping requests

Instructions

1. Write a script `firewall.py`
2. For a minimal configuration, set two definitions to **False**:
 - `ALLOWOUTGOINGICMP`: If set to True, firewall should allow outgoing ping requests (initiated by your computer)
 - `ALLOWINCOMINGICMP`: If set to True, firewall should allow incoming ping requests (sent to your computer)
3. Write the basic PyDiver loop
 1. Sniff with a wide filter ('ip')
 2. Check per packet object, if it's an ICMP request (google "ICMP request types")
 1. If so, check `ALLOWOUTGOINGICMP` or `ALLOWINCOMINGICMP` accordingly, and drop packet if not allowed
 3. Remember, if you're not dropping the packet - you need to send it (`w.send(packet)`)
4. Test the outgoing filtering by trying to send pings
5. Test the incoming filtering by asking a classmate to send you a ping
6. Remember to test your script by changing the configuration: allow outgoing/incoming and try again while the script is running. This is a good way to make sure your configuration is relevant (used properly by the code).
 1. For example, if the configuration allows outgoing but not incoming, you should be able ping from your computer!

To submit

Submit script `firewall.py`.

